RONALD MILGO

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EDUCATION

Yale University, *New Haven, CT* Expected Graduation, May 2027

- BS. Computer Science
- *Relevant Coursework*: Data Structures and Programming Techniques, Algorithms, Full Stack Development, Discrete Math, Multivariable Calculus, Computer Architecture, Object-Oriented Programming, Systems Programming & Computer Organization

WORK EXPERIENCE

Software Engineer Intern **New Haven, CT**
Tsai Center for Innovative Thinking at Yale May 2025 - Present

- Designed and deployed a distributed cloud pipeline for **Neotix Robotics** that ingested 10TB+ of weekly multimodal robotics data, cutting latency by 40% via AWS Lambda, S3, and gRPC microservices; integrated ML-assisted annotation tools that tripled labeling throughput across 100K+ episodes and reduced error rates by 35%.
- Led firmware and backend systems at **Verustruct** for a gantry-less 3D printer, enabling real-time infrastructure monitoring and carbon-negative wall fabrication with sensor-driven accuracy.
- Built a real-time generative AI engine at **Prevision Labs** that transformed text prompts into 50+ FPS visuals for live performances by optimizing CUDA-based inference pipelines with asynchronous batching and multi-threaded GPU streaming—reducing latency by 45% and powering 200+ interactive showcases for VJs and creative teams.

Software and Systems Engineer Intern **New Haven, CT**
Yale School of Engineering & Applied Science May 2024 - August 2024

- Built Python automation tools for lab inventory and scheduling workflows, reducing equipment setup time by 30% and improving lab efficiency for 100+ students and faculty.
- Designed and implemented a predictive scheduling algorithm using Python and historical usage data, forecasting lab equipment demand and reducing resource conflicts by 40% during peak project cycles.
- Programmed Arduino-based engineering bikes, enabling 40+ students to interact with motors, sensors, and embedded systems—facilitating hands-on learning in IoT, robotics, and real-time control.
- Maintained and troubleshot lab software environments using MATLAB, LabVIEW, and hardware interfaces, ensuring uninterrupted operation across 100+ concurrent research projects.

Technology & Software Intern **Nairobi, Kenya**
Equity Bank Limited May 2022 - May 2023

- Implemented a CRM system to track and follow up with new account holders, using **SQL** and workflow automation tools to boost customer retention by 35% in the first 3 months.
- Taught **Python** and **C programming** to 300+ students as part of a nationwide tech outreach initiative, introducing foundational programming concepts and promoting CS majors among high-potential scholars.

TECHNICAL SKILLS

- **Programming languages:** Python, C, C++, C#, JavaScript, Java, Swift, Rust, Go, Ruby, PHP, HTML, CSS, SQL, Racket.
- **Frameworks & Libraries**: Flask, Django, React.js, Next.js, Angular, Vue, Node.js, Spring Boot, RESTful APIs, TensorFlow, PyTorch
- **Tools & Platforms**: AWS, Git, GitHub, Linux, MongoDB

TECHNICAL PROJECTS & LEADERSHIP EXPERIENCES

Web developer **New Haven, CT**
Yale International Students' Organization December 2023 - April 2025

- Developed interactive marketing components for the organization's website using **HTML**, **CSS**, and **JavaScript**, increasing user engagement and page interaction by 30% across event-related content.
- Designed and deployed responsive digital materials for weekly events, optimizing visuals for both email campaigns and Instagram, reaching an audience of 1,000+ students.
- Collaborated with the Marketing Council to improve UI consistency across platforms, enhancing brand cohesion and digital reach.

Web Application
Sorting Algorithm Visualizer November 2024 - January 2025

- Developed an interactive web-based visualizer using **HTML**, **CSS**, and **vanilla JavaScript** to animate sorting algorithms including Merge Sort, Quick Sort, and Bubble Sort, supporting real-time control of speed and array size.
- Created as a learning tool for a **Yale Data Structures course**, helping 200+ classmates better understand algorithmic behavior through live visual demonstrations and in-class walkthroughs.

Full-Stack Web Application
BulldogFit January 2025 - April 2025

- Built a community-focused fitness web app for Yale students using **HTML**, **CSS**, **JavaScript**, and **Flask**, enabling 100+ users to view real-time gym availability and coordinate workouts based on shared schedules and fitness goals.
- Integrated a secure **real-time chat feature** using **WebSockets** and **Flask-SocketIO**, allowing matched users to connect, plan sessions, and build a sense of community around shared fitness interests.